

```

46. # Check for the special case where the start and end points are
47. # certain neighbors, which this function doesn't handle correctly,
48. # and return a hard coded list instead:
49. if (x1 == x2 and y1 == y2 + 1) or (y1 == y2 and x1 == x2 + 1):
50.     return [(x1, y1), (x2, y2)]
51.
52. isSteep = abs(y2 - y1) > abs(x2 - x1)
53. if isSteep:
54.     # This algorithm only handles non-steep lines, so let's change
55.     # the slope to non-steep and change it back later.
56.     x1, y1 = y1, x1 # Swap x1 and y1
57.     x2, y2 = y2, x2 # Swap x2 and y2
58. isReversed = x1 > x2 # True if the line goes right-to-left.
59.
60. if isReversed: # Get the points on the line going right-to-left.
61.     x1, x2 = x2, x1 # Swap x1 and x2
62.     y1, y2 = y2, y1 # Swap y1 and y2
63.
64.     deltax = x2 - x1
65.     deltax = abs(y2 - y1)
66.     extray = int(deltax / 2)
67.     currenty = y2
68.     if y1 < y2:
69.         ydirection = 1
70.     else:
71.         ydirection = -1
72.     # Calculate the y for every x in this line:
73.     for currentx in range(x2, x1 - 1, -1):
74.         if isSteep:
75.             points.append((currenty, currentx))
76.         else:
77.             points.append((currentx, currenty))
78.         extray -= deltax
79.         if extray <= 0: # Only change y once extray <= 0.
80.             currenty -= ydirection
81.             extray += deltax
82. else: # Get the points on the line going left to right.
83.     deltax = x2 - x1
84.     deltax = abs(y2 - y1)
85.     extray = int(deltax / 2)
86.     currenty = y1
87.     if y1 < y2:
88.         ydirection = 1
89.     else:
90.         ydirection = -1
91.     # Calculate the y for every x in this line:
92.     for currentx in range(x1, x2 + 1):
93.         if isSteep:
94.             points.append((currenty, currentx))
95.         else:
96.             points.append((currentx, currenty))
97.         extray -= deltax
98.         if extray < 0: # Only change y once extray < 0.
99.             currenty += ydirection
100.            extray += deltax

```